PJM reserve reforms would be moderately bullish to LMPs if approved

North America Power
20 Mar 2026, GMT | 6 min read

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- **PJM reserve proposal would be bullish LMPs, particularly during summer, winter net load ramping hours due to new reserve products.**
- **ISO proposes new ramping products similar to ERCOT, seeking to mitigate net-load uncertainty as renewable penetration rises.**
- **Total summer day-ahead reserve requirements would rise by up to 0.4 GW (10%) during solar hours; ISO expects LMP uplift.**
- **Partially unnesting reserve products would result in higher maximum system marginal energy price of $4,100/MWh.**
- **Earliest implementation could be peak-winter 2027 given October FERC filing date, but stakeholder approval far from certain.**

The PJM Reserve Certainty Senior Task Force's (RCSTF) 17 March proposal to dramatically overhaul reserve markets—by introducing several new products, associated methodologies to establish requirements and new operating reserve demand curves (ORDC)—would be moderately bullish LMPs if approved. But the future of the proposal is far from certain given its size and complexity, and as the RCSTF's proposal to raise the day-ahead (DA) 30-minute reserve requirement failed to pass the Markets and Reliability Committee (MRC) in 2024. The proposal is bullish LMPs in both the DA and real-time (RT) markets, but the impact would be largely isolated to high net-load days in summer and winter, when prices would already be elevated.

The RCSTF is targeting a June vote to approve the proposal, sending it to the MRC for approval in August and the Members Committee in September. The ISO would then file the stakeholder-approved proposal with FERC in October. PJM has not stated an implementation target for the proposal, but the earliest potential go-live date would likely be 1 January 2027. We are not currently raising our calendar-year 2027 spark spread forecast in PJM due to the uncertainty around approving the proposal and ultimate implementation date, in addition to the price uplift being mostly concentrated on high net-load days—although that could change as PJM solar capacity installations continue.

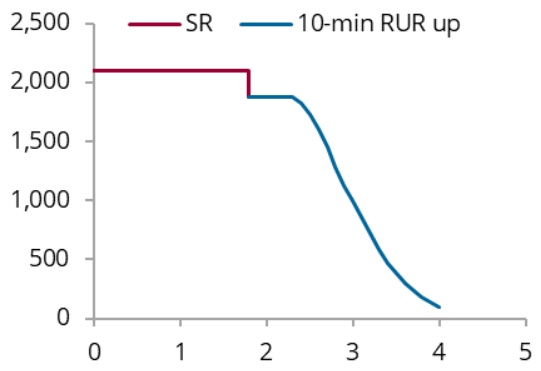
More products, more complexity

In both the DA and RT markets, the ISO proposes to retire the Primary Reserve (PR) service and replace it with Ramp/Uncertainty Reserves (RUR), aimed at addressing increasing net-load ramping uncertainty caused by forecast error for load and renewable generation. The new products are 10-minute RUR up and

down, and 30-minute RUR up. The 30-minute RUR up will be nested within the existing Total 30-minute reserve service, meaning any quantity of 30-minute RUR procured also satisfies the Total 30-minute requirement. In today's reserve markets, Synchronized Reserves (SR), PR and 30-minute reserve are nested, which also means reserve prices and penalty factors stack. Importantly, SR, 10-minute RUR and 30-minute RUR will unnest, creating new ORDCs that, when coupled with new requirement calculations, result in higher reserve procurement and uplift to LMPs through co-optimisation.

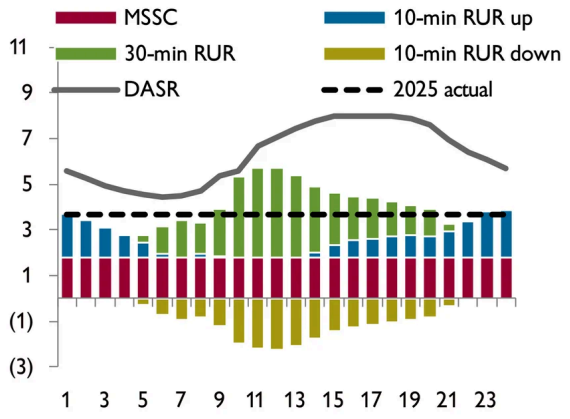
ORDCs under the proposal would take similar shapes to those constructed in ERCOT's Real-Time Co-optimization plus Batteries (RTC+B), which would allow PJM to go short reserves if doing so would result in the least cost solution. In practice, PJM will procure the most valuable reserves first, starting with SR, incrementally satisfying 10-minute RUR next, 30-minute RUR, and finally 30-minute reserve requirements. This creates a hierarchy within the ORDCs such that the maximum willingness to pay for each reserve product, or the top of each ORDC, is weighted by that product's value in the hierarchy. Therefore, the cost to avoid a shortage of the most valuable reserve product, SR, would be the maximum reserve penalty price. The proposal would also raise the administrative energy price cap from $3,700/MWh to $4,100/MWh due to a higher cap on the SR ORDC of $2,100/MWh (see Figure 1). When added to the $2,000/MWh hard energy offer cap, system energy prices could max out $400/MWh higher than the current level, with final LMPs potentially rising further due to congestion and losses.

Fig 1: PJM example 10-min RUR up composite ORDC, $/MWh (y-axis), GW (x-axis)



Note: Indicative requirement values used
Source: PJM, Energy Aspects

Fig 2: 24 June DA simulated, actual reserve requirements, GW



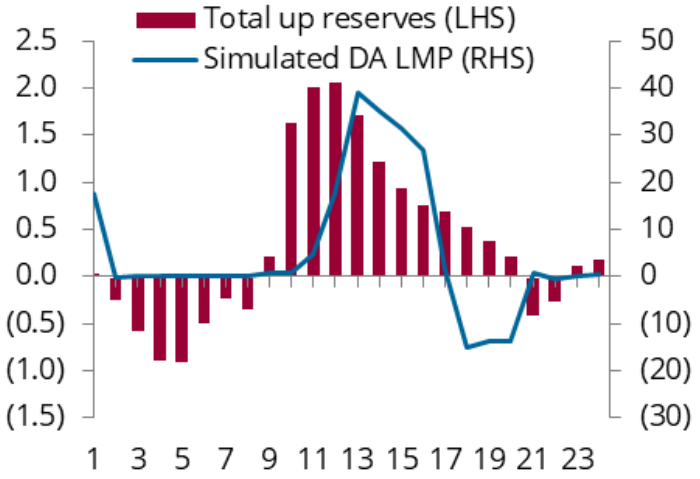
Source: PJM, Energy Aspects

The proposal also aims to bring operator out-of-market commitments, typically made after the day-ahead market clears, into market via the creation of a Day-Ahead Scheduling Reserve (DASR) service, similar to ISONE's Day-Ahead Ancillary Services Initiative (DASI). The DASR reserve requirement represents the total reserve requirement across all up services for a given hourly interval, which can be satisfied by any of the procured RUR up products (see Figure 2). If the DASR requirement is still not met, PJM must procure additional secondary reserves, but these would have a market clearing price cap of $50/MWh and do not clear above $0/MWh outside of rare grid emergencies.

Higher prices

PJM simulated the proposed new reserve requirements for each settlement interval in 2025, and calculated the impact on hourly DA LMPs on several seasonally representative days relative to a backtest that assumed no changes to reserve requirements. The ISO's results show on average $5/MWh uplift to DA around-the-clock (ATC) LMPs on the representative high-load summer days, with the greatest uplift of $6.14/MWh simulated on 24 June (see Figure 3). But DA ATC prices in that day's simulation nearly reached $170/MWh, meaning new reserve requirements would have only raised prices by 3%. The representative high-load winter day also had more than $8/MWh of uplift, while spring and autumn days experienced a slight reduction in LMPs, likely due to lower net loads.

Fig 3: 24 June net change in total reserve requirement, DA LMPs, GW (LHS), $/MWh (RHS)



Source: PJM, Energy Aspects

We estimate total up reserve requirements, inclusive of the Most Severe Single Contingency (the new SR requirement, see Figure 1), 10-minute RUR up and 30-minute RUR up, would have led to an average of 0.7 GW more reserves procured during the on-peak block compared to actual levels in summer 2025. The largest rise in procurement occurred over HE10–14, in which PJM would have procured over 1.7 GW more reserves on average, almost entirely in the 30-minute RUR product. That slice of the on-peak block is when net load ramping is steepest, ultimately realising at nearly 5.9 GW/hour during the operating day last June. The simulation showed flat to slightly higher reserve procurement over the July–August super-peak block of HE17–19, but showed DA LMPs in the sample day falling by $14/MWh in those hours. This could be due to the decline in the 30-minute RUR requirement as the likelihood of the system needing longer duration up-ramping reserves falls at the peak-load hour. Falling LMPs during that block could also be due to new ORDCs that allow for a gradual increase in reserve prices along a smooth ORDC, rather than the existing multi-step curves.

Analysts

New York | Brian Myers, Daniel Pyrek